

When Yield Curves Invert Together: Predicting Currency Carry Losses

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August 2026

Job Market Paper

Abstract

National yield curves mix country-specific and shared information. I use their cross-country configuration to construct a predictive state: fresh 10-year/2-year inversions must be live in at least two non-U.S. G10 economies. From 1988 to 2026, annualized carry-trade spot returns are 12.9 percentage points lower when the lagged state is active, with inference based on fifteen episodes. Separate exactly-one and exactly-two state coefficients have opposite signs, and the configuration retains predictive content alongside the U.S. curve, the average G10 slope, and individual-country states. Losses fall on the high-interest-rate currencies bought by carry trades and grow with predetermined S&P 500 equity beta. The precise carry estimate depends on tenor and release design, while the exposure ordering is more stable. Cross-country yield-curve configurations therefore reveal a predictive state for exposure-ordered currency losses.

JEL: E43, F31, G12, G15.

Keywords: currency carry trade, yield-curve inversion, exchange-rate predictability, international bond markets.

1 Introduction

Currency portfolios diversify national shocks but remain exposed when several currencies move together. This distinction is central to the carry trade. Carry buys a standing set of high-interest-rate currencies and funds it with low-interest-rate currencies. The interest differential accrues gradually; one shared depreciation of the long leg can erase many months of income. This paper asks whether the cross-country configuration of bond markets reveals that loss state before the currency payoff.

One yield-curve inversion is ambiguous. It can reflect domestic weakness, country-specific disinflation, or a national term-premium movement. A cluster of fresh inversions is a different object: several bond markets cross the same economically salient boundary within overlapping windows. I construct a monthly state from 10-year/2-year slopes in nine non-dollar G10 economies. A fresh inversion observed in month $t - 1$ becomes live in month t and remains live until its slope has risen for two consecutive months. The state is active when at least two curves are live. The month- t state predicts the return in month $t + 1$.

The first result is a large conditional spot-return shift. The 1988–2026 sample contains 92 active months in fifteen episodes. Annualized carry spot returns average 2.85 percent outside the state and -10.00 percent inside it, a difference of -12.85 percentage points. The money-market-differential total-return proxy falls from 6.45 to -4.86 percent. The known interest differential remains positive, but it absorbs only part of the depreciation. The location shift reaches beyond the worst observations: the monthly spot median falls from 0.35 to -0.76 percent, and an outcome-conditioned bottom-decile deletion leaves a -7.24 -point spot difference.

The second result isolates the cross-country configuration. Separate indicator regressions give a positive coefficient for exactly one live inversion and a -15.72 -point annualized coefficient for exactly two; they do not provide a direct one-versus-two contrast. In joint regressions, the cluster retains predictive content alongside the U.S. inversion, the continuous average G10 slope and its change, and all nine country-level live states. The raw count and nearby release and tenor rules reveal meaningful sensitivity, not a universal cutoff: the 10Y–3M and three-month-release constructions attenuate the carry coefficient. The empirical object is therefore the nonlinear configuration of fresh crossings, with a precise historical estimate near the baseline rule, rather than “inversion” in general.

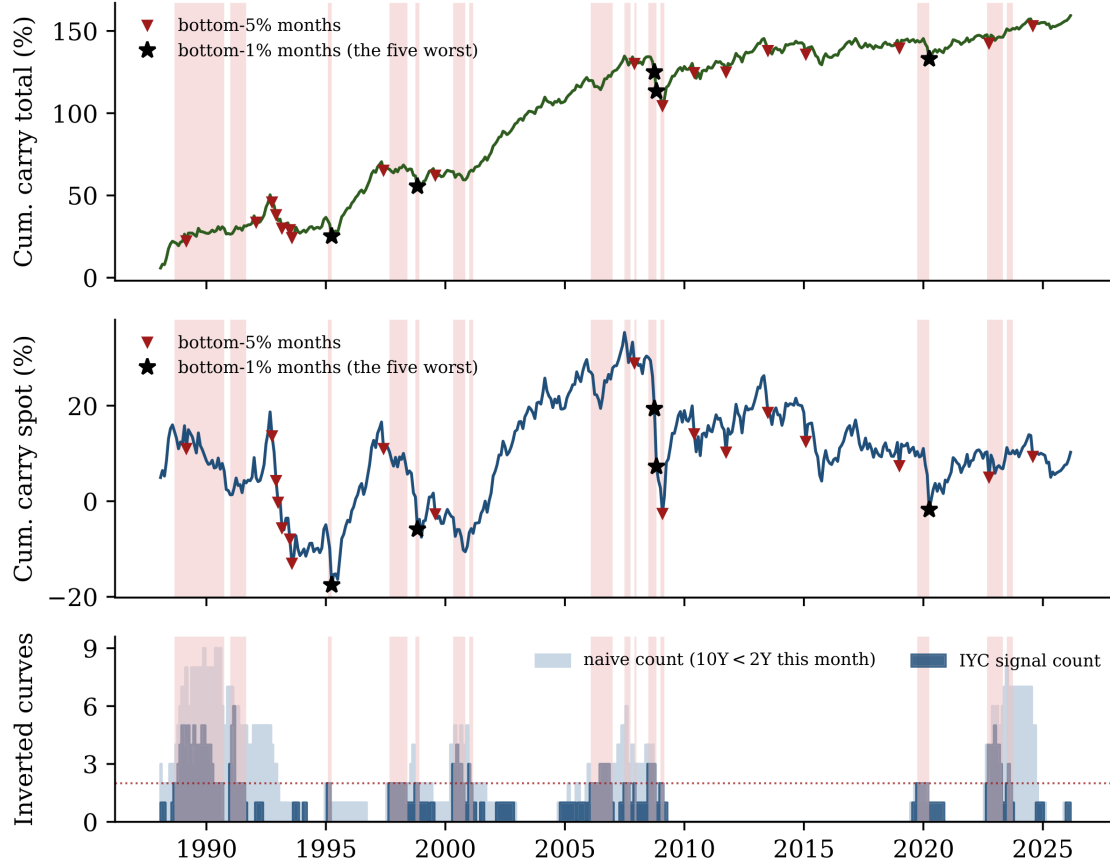


Figure 1: Carry returns and synchronized inversions. The light and dark lines count currently inverted and live curves; the dotted line marks the threshold of two. Shading is the month-end state, used with a one-month lag in return regressions.

The third result shows where the loss lands. High-interest-rate currencies depreciate relative to the low-rate currencies used to fund carry. Bilateral state coefficients also become more negative with currencies' predetermined equity-market betas. The cross-currency gradient explains 77 percent of the variation across nine coefficients; a portfolio sorted directly on expanding-window S&P 500 betas loses 7.53 percentage points per year in spot returns in the state. The exposure result is not uniform across every benchmark: an MSCI World beta sort is weak over the maximal full sample but agrees after 1999. This pattern makes the cross-sectional content sharper than a generic dollar or volatility state while keeping the beta measure itself contestable.

The paper's contribution is to change the unit of observation. National yield slopes forecast activity and combine expected short-rate paths with term premia (Harvey, 1988; Estrella and Hardouvelis, 1991; Estrella and Mishkin, 1998; Campbell and Shiller, 1991; Hamilton and Kim, 2002). International curves share important variation (Plosser and Rouwenhorst, 1994; Bernard and Gerlach, 1998; Diebold et al., 2008; Chinn and Kucko, 2015). Carry returns, in turn, load on common currency and stress factors (Fama, 1984; Brunnermeier et al., 2008; Lustig et al., 2011; Menkhoff et al., 2012; Farhi and Gabaix, 2016). The new empirical object is neither one country's slope nor a smooth global average. It is a cross-country threshold configuration, dated before the payoff and paired with a predetermined exposure gradient.

That configuration produces a disciplined economic interpretation. Write national slopes as a shared component plus country-specific information. Several fresh threshold crossings are more consistent with shared information than one crossing under restrictions on the idiosyncratic components. If currencies differ in their exposure to a subsequent adverse return component, the losses should be ordered across currencies. The exact-one/exact-two contrast, the reported curve-state horse race, and the beta gradient implement those predictions. They support a shared-information interpretation; they do not distinguish global growth, coordinated policy, inflation, term premia, or intermediary risk-bearing as the primitive shock.

Two design facts govern the strength of the claim. First, fifteen episodes supply the effective common-state variation. Second, the tenor, threshold, freshness rule, and release rule were assessed on the same historical sample. I therefore show the reported local state-definition neighborhood and add an independently executable public 10Y–3M audit that exhaustively reports a declared 64-rule family. The public family frequently yields negative coefficients, but no rule meets the 5-percent common-

calendar rotation-reference criterion. That adverse result places the ambition of the paper where the evidence is strongest: a novel cross-country configuration and exposure pattern, not an immaculate universal trading rule.

A public state that forecasts a negative total-return proxy also raises a pricing question. Predetermined carry compensates exposure only if it adjusts enough before the loss. A compact benchmark shows the restriction required for current expected losses to exceed both ordinary income and previously embedded compensation. It does not identify slow adjustment. The empirical contribution is prediction and exposure ordering; the friction remains an open mechanism.

Section 2 develops the framework, return accounting, state, and data. Section 3 defines estimation and inference. Section 4 establishes the cross-country configuration, and Section 5 shows where losses fall. Section 6 examines alternative explanations and the independent public audit. Section 7 states the public-information puzzle. Section 8 concludes. The online appendix gives the exact recursion, derivations, supplementary evidence and adverse robustness, and executable public-data checks.

2 Economic Framework and Measurement

2.1 Cross-country configurations as information

Let the 10-year/2-year slope in country i be

$$q_{i,t} = \mu_i + f_t + u_{i,t}, \quad (1)$$

where μ_i is a country level, f_t is information shared across curves, and $u_{i,t}$ collects country-specific information and measurement error. A lower f_t tends to flatten several curves. The inversion indicator is

$$I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}. \quad (2)$$

One threshold crossing can arise from either component. Several fresh crossings are harder to attribute to one country's $u_{i,t}$. This signal-extraction logic requires restrictions on the cross-country dependence of the country-specific components: regional shocks, coordinated policy, or common measurement errors can also produce synchronization. Equations (1)–(2) therefore motivate empirical comparisons; they do not identify f_t or assign a structural posterior probability to two crossings.

The framework yields a nonlinear prediction. Exactly one inversion need not be a weaker adverse state than two. A single curve can identify a national rate cycle, whereas a cluster can change the composition of likely information. It also yields a spanning prediction: if the cross-country configuration matters, it should add information beyond one country's curve and a continuous average slope. The threshold and path rules are empirical measurements of that configuration, not primitives of the model.

The yield-slope component and the subsequent currency-return innovation are distinct. Write

$$\Delta s_{i,t+1} = -\lambda_i h_{t+1} + e_{i,t+1}, \quad (3)$$

where $h_{t+1} > 0$ is an adverse return component shared across currencies, λ_i is currency i 's exposure, and $e_{i,t+1}$ is idiosyncratic. The synchronized state is informative when $\mathbb{E}[h_{t+1} \mid S_t = 1]$ exceeds its inactive-state counterpart. Equation (3) then predicts larger depreciations for high- λ_i currencies. Predetermined equity beta is the empirical exposure proxy. The paper does not equate f_t with h_{t+1} ; the first is information in bond-market configurations, and the second organizes the currency payoff.

2.2 Currency returns and the shared-loss problem

Consider a U.S. investor holding currency i . Let $s_{i,t}$ be the log dollar price of one unit of foreign currency, so $\Delta s_{i,t+1} > 0$ is foreign-currency appreciation. Let $d_{i,t} \equiv (i_{i,t} - i_{US,t})/12$ be the one-month interest differential, formed from annualized money-market rates. The excess-return proxy is

$$rx_{i,t+1} = d_{i,t} + \Delta s_{i,t+1}. \quad (4)$$

The differential is known at t ; the spot change is realized from t to $t + 1$. Under covered interest parity this sum corresponds approximately to a forward return apart from basis and measurement differences (Du et al., 2018). It combines a simple annualized interest-rate differential divided by twelve with a log spot change, a first-order monthly approximation. Because the paper uses money-market differentials rather than executable forward quotes, the object is an asset-pricing return proxy, not a transaction-cost-adjusted strategy.

Each month, the carry portfolio buys the three highest-rate currencies and sells the three lowest-rate currencies among the nine non-dollar G10 currencies. If $\mathcal{H}_t$ and $\mathcal{F}_t$ denote the high-rate and funding sets, then

$$\begin{aligned} rx_{t+1}^{\text{carry}} &= \frac{1}{3} \sum_{i \in \mathcal{H}_t} rx_{i,t+1} - \frac{1}{3} \sum_{i \in \mathcal{F}_t} rx_{i,t+1} \\ &\equiv C_t + X_{t+1}. \end{aligned} \quad (5)$$

Here C_t is the long-minus-short interest differential and X_{t+1} is the corresponding spot return. The sample-average income leg is 3.9 percent per year and is positive in every month, with a minimum of 0.7 percent. Carry therefore loses only when the spot leg is more negative than the income leg is positive.

Diversification reduces $e_{i,t+1}$ but not the shared component when the long and funding legs have different average exposures. Here “shared” describes cross-currency comovement; “international” is the economic interpretation. The return equation is an organizing device rather than an estimated factor model.

2.3 The synchronized-inversion state

Let $q_{i,t} \equiv y_{i,t}^{10Y} - y_{i,t}^{2Y}$ denote country i 's end-of-month yield slope. An inversion can reflect expected easing, term-premium variation, or both (Campbell and Shiller, 1991; Hamilton and Kim, 2002; Bauer and Mertens, 2018).

Define $I_{i,t} = \mathbf{1}\{q_{i,t} < 0\}$. A crossing observed at $t - 1$, $I_{i,t-2} = 0$ and $I_{i,t-1} = 1$, makes the curve live at t . It remains live until $q_{i,t}$ has increased for two consecutive months. A released curve cannot re-enter while still inverted; it must first become nonnegative and then cross below zero again. Let $L_{i,t}$ be the resulting live status. The central state is

$$N_t = \sum_{i=1}^9 L_{i,t}, \quad S_t = \mathbf{1}\{N_t \geq 2\}. \quad (6)$$

Thus $L_{i,t}$ is path dependent, whereas $I_{i,t}$ depends only on the current slope. The state measured at the end of month t conditions the return during month $t + 1$.

The live rule has two economic roles. Delayed fresh entry excludes inversions already embedded in prices for many months and prevents the crossing-month return from entering the predictive state. Confirmed release allows the state to remain active as curves re-steepen, which can coincide with delivered easing and delayed currency adjustment. In the sample, a raw two-or-more current-inversion state is active in 165 months, compared with 92 for the live state, yet switches off before March 1995, October 2008, and March 2020—the three largest carry-loss months. This historical fit is informative and also creates specification-search risk. Online Appendix A states the transition rules and boundary cases exactly.

2.4 Data, information set, and predetermined exposure

The monthly sample runs from January 1988 through February 2026. The currency universe comprises the Australian dollar, Canadian dollar, Swiss franc, euro (Deutsche mark before 1999), British pound, Japanese yen, Norwegian krone, New Zealand dollar, and Swedish krona. The dollar is the numeraire and does not enter either the sort or the baseline state count. The U.S. curve is retained as a benchmark.

State incidence varies across monetary regimes. It is active in 33 of 84 months during 1988–94 but only 3 of 120 months during 2010–19, when many policy rates were near their lower bounds. The state is not a continuously available factor; it appears when international rate cycles create enough cross-country slope variation. Online Appendix

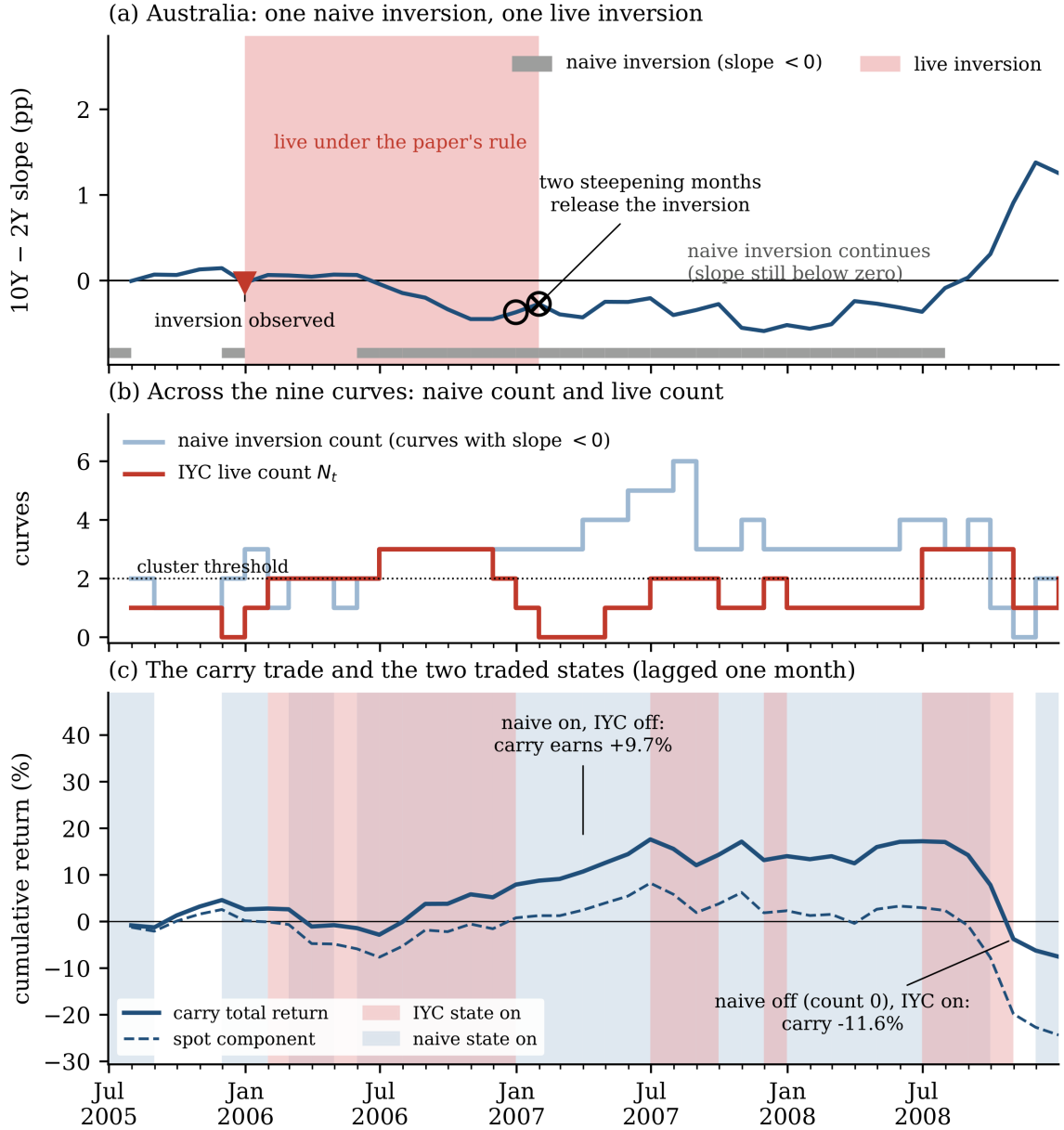


Figure 2: Live-state construction, 2005:07–2008:12. Panel (a) shows the Australian 10Y–2Y slope and its live status, which begins one month after the crossing is observed. Panel (b) compares the current-inversion and live counts. Panel (c) plots cumulative carry returns under the two lagged states. Delayed fresh entry removes the crossing-month return and stale inversions; confirmed release can retain the state after the current-inversion count declines.

A reports the full era counts.

The baseline state uses end-of-month 2-year and 10-year government yields. Portfolio weights use one-month money-market rates known at t . Return regressions use no information dated after t to predict month $t + 1$. This convention prevents mechanical look-ahead but does not eliminate asynchronous market closes, stale quotes, or later data revisions.

The exposure test estimates a currency’s equity beta on an expanding window:

$$rx_{i,\tau} = a_i + \hat{\beta}_{i,t} r_\tau^M + u_{i,\tau}, \quad \tau \leq t - 1. \quad (7)$$

The estimate $\hat{\beta}_{i,t}$ is available before the return from t to $t + 1$ and requires at least 36 prior monthly observations. The one-month buffer is part of the baseline construction: the estimate paired with S_t uses currency and equity returns only through $t - 1$. The beta is an empirical proxy for the latent exposure λ_i in equation (3); it need not equal that primitive. S&P 500 total returns form the market measure.

Table 1: Notation and timing

Symbol	Definition	Timing
$t; t + 1$	Information month-end; subsequent holding month	Information $\rightarrow$ return
$s_{i,t}; \Delta s_{i,t+1}$	Log dollar price of currency i ; its one-month change	Spot change realized in $t + 1$
$d_{i,t}$	Monthly foreign-minus-U.S. interest differential	Known at t
$rx_{i,t+1}$	Currency excess-return proxy	Realized in $t + 1$
$C_t; X_{t+1}$	Carry income leg; carry spot leg	Known at t ; realized in $t + 1$
$q_{i,t}$	Country i ’s 10Y–2Y yield slope	Observed at month-end t
$I_{i,t}; L_{i,t}$	Current inversion; path-dependent live inversion	Information at t
$N_t; S_t$	Live-curve count; synchronized-inversion state	Predict $t + 1$
$f_t; h_{t+1}$	Shared slope information; adverse currency-return component	Information at t ; payoff in $t + 1$
$\lambda_i; \hat{\beta}_{i,t}$	Latent return exposure; estimated equity-beta proxy	Beta uses data through $t - 1$
W_t	Vector of lagged benchmarks or controls	Known at t

Notes: Equations use the information-time convention $t \rightarrow t + 1$. Tables indexed by the return month equivalently report S_{t-1} .

3 Empirical Design

3.1 Predictive regression and inference

All return tests use one timing convention: information at t predicts the return at $t + 1$. The general specification is

$$Y_{t+1} = \alpha + bS_t + \Gamma'W_t + \varepsilon_{t+1}, \quad (8)$$

where W_t contains the relevant lagged benchmarks or controls. The headline conditional-mean difference sets W_t empty. With a binary state, b is the active-minus-inactive mean difference; monthly return coefficients are multiplied by twelve for reporting. In tables indexed by return month, equation (8) is equivalently written with S_{t-1} .

The state’s 92 active months form fifteen contiguous episodes, fourteen longer than one month. Treating all active months as independent would overstate the information in the sample. I report Newey–West statistics with 12 lags and, for binary-state tests, a wild episode bootstrap that assigns Rademacher weights to contiguous episodes and imposes the null. The episode bootstrap is the headline inference because continuation months do not create independent synchronized shocks. The estimates can be economically large while their independent common-state variation remains fifteen episodes.

Distributional diagnostics report medians, lower quantiles, skewness, expected shortfall, and conditional variance. A separate diagnostic removes each outcome’s full-sample bottom-decile observations from both states before re-estimating equation (8). Because that deletion conditions on realized returns, it is not a preferred estimator. It answers only whether a handful of extreme observations mechanically generate the mean gap.

3.2 Benchmarks and controls

The benchmark hierarchy moves from nearby measurement alternatives to broader risk states. The raw current-inversion state asks whether fresh entry and confirmed release add information. The U.S. inversion asks whether the conventional single-country proxy spans the international state. The average G10 slope and its change ask whether the threshold is merely a nonlinear transformation of a smooth global slope. Nine country-specific live states ask whether synchronized crossings add information beyond

their individual components.

Lagged NFCI and VIX controls address predictive spanning: do variables already known at t absorb the state’s forecast? Contemporaneous dollar and currency-volatility factors answer a different question: do those realized factors account for the loss after it occurs? Neither design identifies the shock represented by S_t . Joint specifications are reported without sequential orthogonalization because the ordering of closely related yield-curve transformations would otherwise determine which variable receives their shared variation.

3.3 Exposure ordering

The bilateral state coefficients come from

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}. \quad (9)$$

The nine estimates $\hat{b}_i$ are compared with currencies’ average predetermined equity betas. The slope, fitted intercept at zero beta, and uncertainty describe the exposure gradient. A beta-sorted portfolio and a panel interaction $S_t \times \hat{\beta}_{i,t}$ provide complementary tests: the portfolio is economically transparent; the panel uses monthly variation but still has only fifteen common-state episodes.

This design separates three claims. Prediction asks whether S_t forecasts returns. Exposure accounting asks whether the realized losses are ordered by a predetermined market characteristic. Mechanism identification would require independent variation that distinguishes global growth, policy, term-premium, inflation, and intermediary channels. The first two are directly estimated; the third is not.

3.4 Specification-family discipline

Two different specification exercises appear in the paper and are never pooled. The first is the state-definition neighborhood reported for the baseline 10Y–2Y analysis. It varies breadth, release confirmation, tenor, and current-count repairs. These rows reveal where the historical result strengthens or attenuates, but they are selected displayed specifications rather than the complete set of rules examined. Their p -values are per specification; the joint resamples required for family adjustment are unavailable.

The second exercise is an independently executable public proxy. Its code declares

and exhaustively reports 64 combinations of raw or live 10Y–3M states, breadth thresholds, release lengths, one- or two-month return lags, and target leg sizes of two or three, with all currencies tied at a cutoff included and equally weighted. It reports every coefficient, an inclusive circular-shift reference, and a common-calendar maximum-absolute-standardized-coefficient reference across the family. Under the maintained null that simultaneous cyclic shifts of the states relative to returns are exchangeable, this construction controls the declared family; without that assumption it is a finite rotation-reference diagnostic. The declaration was not preregistered before the data were examined. The public inputs and outcome differ from the baseline analysis, so this exercise measures sign prevalence and selection sensitivity rather than reproducing the headline estimate.

The evidentiary hierarchy follows from those designs. The baseline state supplies the economic result and cross-sectional ordering. Its displayed neighborhood shows known sensitivity. The public family shows what happens when a nearby declared rule family is exhaustively reported and jointly evaluated. A future observation under a dated, unchanged 10Y–2Y rule remains the clean prospective test.

3.5 Claims and rejection patterns

Four findings would run against the paper’s interpretation: the distribution shift vanishing outside a few tail observations; the U.S. or average-slope measures spanning the configuration; no cross-sectional relation between losses and predetermined exposure; or a public nearby-rule family with no sign resemblance. The reported tests run against the first two patterns. The S&P-based exposure tests run against the third, although the nine-point fit lacks a displayed confidence region, the panel interaction is imprecise, and the full-sample MSCI sort fails. The public evidence is weaker: negative coefficients are frequent, but no rule meets the 5-percent common-calendar rotation-reference criterion. That result preserves the cross-country-configuration question while ruling out a claim that the exact carry coefficient is invariant to reasonable measurement choices.

4 The Cross-Country Configuration

4.1 Conditional carry returns

Carry spot returns are 12.85 percentage points per year lower when the lagged synchronized-inversion state is active. They average 2.85 percent in 366 inactive months and -10.00 percent in 92 active months. The money-market-differential total-return proxy falls from 6.45 to -4.86 percent, a difference of -11.31 points. Episode-bootstrap p -values are 0.001 and 0.003.

Table 2: The carry trade in and out of the IYC signal

Portfolio component	Annualized mean		State regression	
	No signal	Signal	Difference	Episode wild p
Carry spot	2.85	-10.00	-12.85	0.001
Carry total	6.45	-4.86	-11.31	0.003
High-beta spot	1.26	-6.27	-7.53	0.010
High-beta total	3.18	-3.64	-6.82	0.013

Notes: Returns are annualized percentages. The no-signal state contains 366 months and the signal state 92 months over 1988:01–2026:02. The difference is the coefficient on the one-month-lagged IYC-state indicator. Wild-bootstrap p -values cluster by the state’s fifteen contiguous episodes ($B = 9,999$).

The accounting locates the result. The carry income leg averages 3.91 percent per year and is positive in every month, with a minimum of 0.7 percent. It does not fall enough to create the conditional loss; the spot depreciation overwhelms it. The object is therefore a predictable spot-risk state and, under the money-market accounting, a negative conditional total-return proxy. Executable forward quotes, basis, spreads, and financing costs are not observed over the full sample.

The distribution shifts with the mean. The monthly carry-spot median changes from 0.35 percent outside the state to -0.76 percent inside it. The 25th percentile moves from -0.93 to -1.98 percent and the 10th percentile from -2.35 to -3.94 percent. Expected shortfall worsens from -4.80 to -8.14 percent. The conditional variance ratio is 1.74, with permutation $p = 0.14$: the evidence is stronger for a leftward shift than for a general variance increase.

Deleting each series’ full-sample bottom-decile months leaves carry spot and total differences of -7.24 and -5.50 points, with episode-bootstrap $p = 0.006$ and 0.024. This deletion uses realized outcomes and is not a preferred estimator. Its narrower

purpose is clear: a small set of worst observations does not arithmetically generate the entire conditional mean.

4.2 Exactly-one and exactly-two indicator regressions

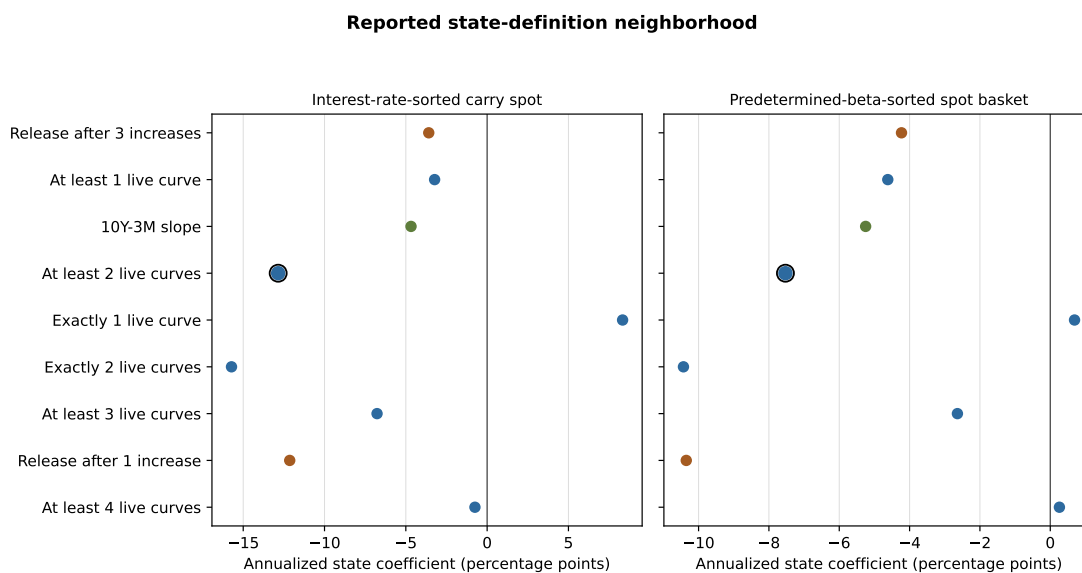
The most revealing comparison is inside the count. In 88 months with exactly one live inversion, carry spot and total coefficients are +8.33 and +8.78 percentage points per year, with $p = 0.005$ and 0.003. In 48 months with exactly two, the coefficients reverse to -15.72 and -15.13 , with $p = 0.001$ and 0.002. The S&P-beta portfolio shows the same qualitative break: +0.69 after exactly one and -10.43 after exactly two.

The opposite coefficient signs are the economic content of synchronization. A lone fresh inversion can arise from one country’s rate cycle, while a diversified carry portfolio continues to collect the cross-country interest differential. At the second curve, the measured configuration changes. Because each coefficient compares its indicated months with a different complement, these regressions do not estimate a direct exactly-one-versus-exactly-two contrast. They nevertheless show why an “any inversion” label can combine return associations with opposite signs. The estimates do not make two a universal structural threshold.

Counts above two do not produce a monotone dose response. At least three live curves gives a -6.77 -point carry-spot coefficient, and at least four gives -0.75 ; both are imprecise and use fewer months. These states also occur later within episodes. The nested at-least-two rule retains the configuration after additional curves enter; the exact-two row is the sharper historical cell, not a stable rule that switches off when a third curve inverts.

4.3 The reported state-definition neighborhood

Figure 3 displays the nine unique breadth, release, and tenor rules reported in the analysis. Rules are ordered by active-month count, independently of their return coefficient. Eight of nine carry-spot estimates and seven of nine S&P-beta spot estimates are negative. The baseline is the second most negative carry rule and the third most negative beta rule. It is not the single best displayed specification.



Rules are ordered by active-month count, not by estimated return. The black ring marks the baseline.
The displayed specifications do not exhaust the search and do not use multiplicity-adjusted inference.

Figure 3: Reported state-definition neighborhood. Points show the displayed breadth, release, and tenor specifications and are ordered by active months. The black ring marks the 10Y-2Y, at-least-two, two-month-release baseline. The display is not the complete set of rules examined and does not provide multiplicity-adjusted inference.

Table 3: Reported state-definition neighborhood

Rule	Active months	Carry spot b [p]	Carry total b [p]	High-beta spot b [p]	High-beta total b [p]
<i>Panel A: number of live inversions</i>					
≥ 1 (any live inversion)	180	-3.23 [0.230]	-1.90 [0.513]	-4.62 [0.008]	-4.16 [0.014]
$= 1$ (exactly one)	88	+8.33 [0.005]	+8.78 [0.003]	+0.69 [0.818]	+0.66 [0.831]
≥ 2 (baseline)	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
$= 2$ (exactly two)	48	-15.72 [0.001]	-15.13 [0.002]	-10.43 [0.012]	-10.31 [0.011]
≥ 3	44	-6.77 [0.405]	-4.57 [0.694]	-2.64 [0.478]	-1.47 [0.711]
≥ 4	22	-0.75 [0.879]	+2.10 [0.672]	+0.26 [0.967]	+1.87 [0.736]
<i>Panel B: release confirmation</i>					
One steepening month	39	-12.14 [0.059]	-10.49 [0.112]	-10.35 [0.035]	-9.39 [0.051]
Two consecutive months (baseline)	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
Three consecutive months	191	-3.59 [0.161]	-2.11 [0.491]	-4.23 [0.001]	-3.58 [0.017]
<i>Panel C: tenor and downcurve filter</i>					
10Y-2Y baseline	92	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
10Y-3M	107	-4.68 [0.146]	-4.02 [0.260]	-5.25 [0.011]	-5.00 [0.034]
10Y-2Y, excluding downcurve	57	-18.07 [0.001]	-17.61 [0.002]	-12.89 [< 0.001]	-12.59 [0.001]
10Y-3M, excluding downcurve	74	-5.73 [0.177]	-6.10 [0.201]	-7.56 [0.003]	-7.94 [0.004]
<i>Panel D: current-count repairs</i>					
Current count ≥ 2	—	-3.18 [0.047]	-1.06 [0.670]	-4.73 [0.005]	-4.06 [0.012]
Current count, excluding downcurve	—	-3.31 [0.238]	-1.73 [0.581]	-6.43 [0.003]	-6.13 [0.006]
Current count, reciprocal-age weighted	—	-10.43 [0.040]	-9.75 [0.057]	-14.87 [0.001]	-14.46 [0.002]
Live state (baseline)	—	-12.85 [0.001]	-11.31 [0.003]	-7.53 [0.010]	-6.82 [0.013]
Live state, excluding downcurve	—	-18.07 [0.001]	-17.61 [0.002]	-12.89 [< 0.001]	-12.59 [0.001]

Notes: Annualized active-minus-inactive coefficients with the wild-bootstrap p -value on each rule's episodes in brackets; 1988:01–2026:02. These are the displayed breadth, release, tenor, and current-count variants, not a complete record of researcher discretion; search-adjusted or family-wise p -values are unavailable. The two tenor states overlap in 58 months. The 10Y–3M downcurve stay-set has carry total +1.59 with $p = 0.371$. Under NFCI/VIX controls, the current count has carry spot -1.91 [0.576] and total -0.63 [0.847]; the reciprocal-age rule has carry spot -7.55 [0.084] and high-beta spot -13.82 [0.008]; and the downcurve-filtered current count has carry spot -3.99 [0.269] and high-beta spot -7.04 [0.035]. Active-month counts are not displayed for Panel D and are left blank.

The carry estimate has two load-bearing measurement margins. Releasing after one increase leaves a large -12.14 -point spot estimate but less episode-level precision ($p = 0.059$); releasing after three increases expands the state to 191 months and attenuates the estimate to -3.59 ($p = 0.161$). Replacing the 2-year yield with a 3-month rate produces -4.68 ($p = 0.146$). The S&P-beta spot estimates remain negative and statistically detectable under both perturbations. The state is therefore more successful at locating an exposure-ordered currency condition than at delivering an invariant carry magnitude.

The raw current-inversion count supplies a second benchmark. Its carry spot coefficient is -3.18 ($p = 0.047$), compared with -12.85 for the live state. Weighting current inversions by reciprocal age revives the coefficient to -10.43 ($p = 0.040$) and produces a -14.87 S&P-beta estimate ($p = 0.001$). Freshness contains information. The live release rule adds a different timing margin by remaining active as curves begin to re-steepen; that margin also retains several historically large loss months. This is

exactly why the public rule-family audit in Section 6 matters.

4.4 Synchronization versus nearby curve states

The portfolio horse race tests whether the configuration adds information beyond the displayed alternatives. The first joint specification includes the continuous average G10 10Y–2Y slope, its one-month change, and the U.S. inversion. The synchronized-state coefficients remain -13.45 for carry spot and -12.44 for total returns, with episode-bootstrap $p = 0.003$ and 0.012 . The average-slope variables are imprecise in the carry columns, and the U.S. inversion enters positively and imprecisely.

Adding all nine country-level live states makes the synchronized coefficient more negative: -26.79 for carry spot and -27.52 for total returns. The nine own states are jointly informative as well. The larger cluster coefficient is not a causal purification. The related regressors divide variation from the same curves, and the own states absorb many benign single-country months. The appropriate conclusion is that the cross-country configuration contains information not spanned by the reported continuous, U.S., or own-state transformations.

Table 4: The synchronized configuration versus reported portfolio-level rivals

Regressor	Specification (2)			Specification (3)		
	b	t_{NW}	Wild p	b	t_{NW}	Wild p
<i>Panel A: carry spot</i>						
Cluster	−13.45	−3.15	0.003	−26.79	−4.88	< 0.001
Global slope	+0.21	+0.08	—	+4.68	+1.28	—
Δ global slope (1m)	−14.97	−1.15	—	−16.10	−1.23	—
U.S. inversion	+2.95	+0.74	0.465	+2.23	+0.51	0.605
Nine own signals (joint p)	—	—	—	—	—	0.005
<i>Panel B: carry total</i>						
Cluster	−12.44	−2.86	0.012	−27.52	−4.92	< 0.001
Global slope	−0.64	−0.22	—	+4.30	+1.17	—
Δ global slope (1m)	−15.14	−1.17	—	−16.22	−1.24	—
U.S. inversion	+2.42	+0.60	0.581	+1.63	+0.37	0.696
Nine own signals (joint p)	—	—	—	—	—	< 0.001
<i>Panel C: high-beta spot</i>						
Cluster	−6.16	−1.72	0.102	−11.99	−2.02	0.040
Global slope	+2.40	+0.90	—	+5.45	+1.58	—
Δ global slope (1m)	+11.87	+0.92	—	+10.85	+0.87	—
U.S. inversion	+1.81	+0.72	0.450	+1.27	+0.38	0.667
Nine own signals (joint p)	—	—	—	—	—	0.009
<i>Panel D: high-beta total</i>						
Cluster	−5.66	−1.56	0.143	−12.37	−2.07	0.036
Global slope	+2.40	+0.88	—	+5.67	+1.62	—
Δ global slope (1m)	+11.41	+0.89	—	+10.32	+0.83	—
U.S. inversion	+2.57	+1.00	0.308	+2.00	+0.57	0.543
Nine own signals (joint p)	—	—	—	—	—	0.004

Notes: Sample: 1988:01–2026:02. Specification (2) includes the cluster, the average G10 10Y–2Y slope, its one-month change, and the lagged U.S. inversion. Specification (3) adds all nine country-level live states. State coefficients are annualized percentage points; slope coefficients are annualized percentage points per percentage-point slope. Continuous regressors report Newey–West statistics only. Binary-state p -values use a wild bootstrap on the relevant state’s episodes, with other regressors partialled under the null. The nine own-state coefficients are not displayed individually; their joint p -value is shown. Cluster-alone coefficients appear in Table 2. U.S. inversion alone: carry spot −1.90 [0.484], carry total −1.34 [0.637], high-beta spot −3.01 [0.205], and high-beta total −2.03 [0.345].

The U.S. inversion alone reinforces the distinction. It is active in 57 lagged months and has carry spot and total coefficients of −1.90 ($p = 0.484$) and −1.34 ($p = 0.637$). Adding the U.S. curve to the cluster count, however, leaves the baseline result nearly unchanged. The U.S. curve is neither a substitute for the international configuration nor a reason to treat the United States as economically irrelevant.

5 Where the Losses Fall

5.1 Interest rates and predetermined equity exposure

The carry result contains a cross-sectional ordering before any beta is estimated. The portfolio loses because the high-interest-rate currencies it buys depreciate relative to the low-interest-rate currencies used for funding. The bilateral regression

$$\Delta s_{i,t+1} = a_i + b_i S_t + e_{i,t+1}$$

asks whether a second predetermined characteristic orders the nine currency coefficients.

Australia and New Zealand have average S&P equity betas of 0.20 and 0.13 and annualized state spot coefficients of -13.21 and -13.28 percentage points. Japan and Switzerland have betas near zero and coefficients of $+2.35$ and -0.89 . Regressing the nine state coefficients on average predetermined beta produces a slope near -60 percentage points per beta unit, an intercept of -0.75 , and $R^2 = 0.77$.

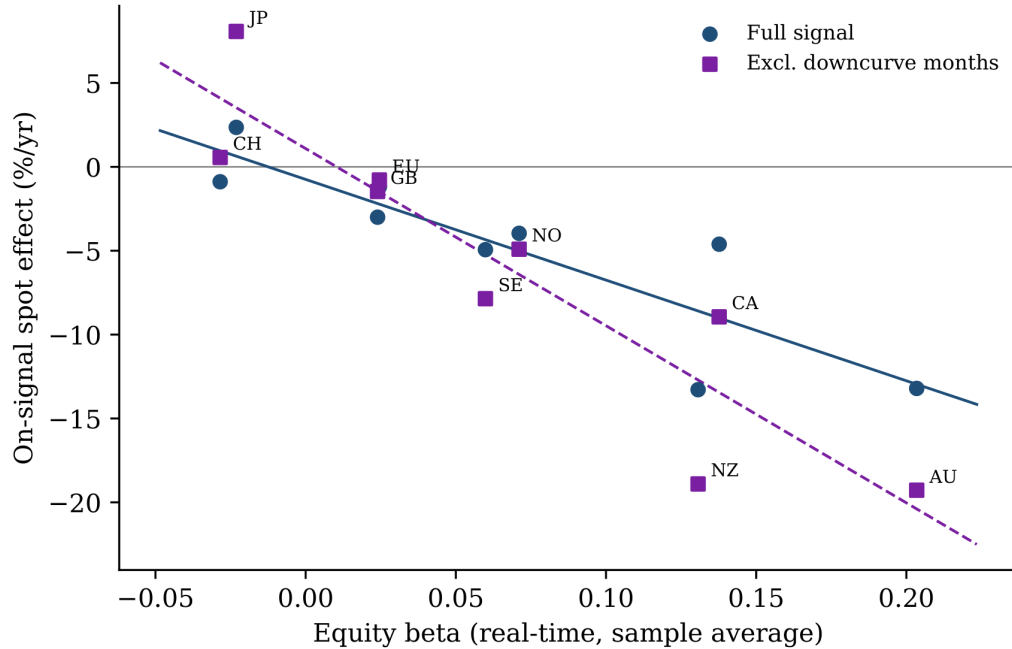


Figure 4: Currency state coefficients and predetermined equity beta. Each point is one bilateral spot-return coefficient. The solid line is the baseline fit (slope -60 , intercept -0.75 , $R^2 = 0.77$); the dashed fit excludes the exploratory downcurve months.

The nine-point gradient is economically sharp and statistically limited. It has no

displayed confidence region for the fitted intercept, and Australia and New Zealand exert substantial leverage. It does not establish a structural beta or prove that exposure vanishes at zero. It documents a descriptive restriction that a uniform dollar shift does not match: in the S&P-based ranking, currencies with greater predetermined market exposure have larger state-conditioned losses.

Table 5: How active-state losses align with exposure

Object	Estimate	Episode wild p	Interpretation
Rate-beta rank correlation	0.72	—	Descriptive, nine currencies
Cross-currency beta slope	−60	—	Percentage points per beta unit
Cross-currency fitted intercept	−0.75	—	Percentage points at zero beta
Cross-currency fit	$R^2 = 0.77$	—	Descriptive, nine currencies
Beta-sorted spot portfolio	−7.53	0.010	Annualized state difference
Panel state $\times$ beta	−27	0.210	Annualized interaction
Conditional Fama slope, active state	−1.33	0.340	p -value is for the state interaction

Notes: The cross-currency slope, intercept, and R^2 come from regressing the nine bilateral state coefficients on currencies' sample-average predetermined equity betas. The beta-sorted portfolio uses expanding-window betas. The panel interaction uses monthly predetermined betas. The Fama row is supporting conditional-moment evidence; its episode-level uncertainty is wide.

The beta-sorted portfolio supplies a portfolio-level implementation of that restriction. It buys the three currencies with the highest expanding-window betas and sells the three lowest, using estimates available before the return. Its spot return is 7.53 percentage points per year lower in the state, with episode-bootstrap $p = 0.010$. The monthly panel interaction between the state and predetermined beta is also negative, about −27 points per beta unit, but its episode-bootstrap p -value is 0.210. The portfolio and bilateral gradient carry the exposure evidence; the panel interaction shows that its precision should not be inferred from the number of currency-month observations.

5.2 The cluster and each currency's own curve

Country-level regressions compare four objects: a currency's own live state alone, the synchronized cluster alone, both together, and a full specification adding own-slope variables and the average G10 slope. The cluster in these regressions still includes the

predicted country’s curve. This is an own-curve horse race, not a leave-one-country-out design.

The Australian own state is the only own-state coefficient detected on its own: -8.76 points with $p = 0.044$. Conditional on the cluster, every own-state coefficient is imprecise. Cluster-alone estimates are -13.21 for Australia ($p = 0.015$) and -13.28 for New Zealand ($p = 0.011$). In the full specification their point estimates remain negative, at -12.92 and -8.56 , but episode-bootstrap precision weakens. Across the pooled bilateral panel, the cluster coefficient is -4.74 with Newey–West $t = -2.34$, while the pooled own-state coefficient is -0.01 with $t = -0.01$.

Own-curve controls are jointly detected for Japan and not for the other eight currencies. This is an economically useful exception: the bundle of Japanese own-state, relative-slope, and relative-level controls is jointly detected even though the yen’s cluster coefficient is positive. The joint test does not isolate which own-curve component supplies the information. More generally, the own-curve specifications do not reproduce the adverse long-minus-short portfolio pattern. A stronger claim—that other countries’ curves predict a currency after its own curve is excluded—requires the unavailable country-month 10Y–2Y state matrix.

5.3 Persistent portfolio cores with some rotation

The rate and beta sorts hold persistent currency blocs. New Zealand occupies the carry long leg in 85 percent of all months and 91 percent of active-state months; Australia does so in 71 and 73 percent. Japan is in the funding leg in 83 percent of all months and 97 percent of active months; Switzerland’s corresponding shares are 96 and 100 percent. For the beta sort, Canada is always in the long leg, Australia is long in 97 percent of all months and 92 percent of state months, and New Zealand in 77 and 76 percent.

The state-conditioned portfolios retain persistent cores, so the result is not created solely by resorting when the signal turns on. There is meaningful rotation around those cores: Norway’s carry-long share falls from 62 percent overall to 34 percent in state months, while the pound’s rises from 43 to 70 percent. Interest rates matter twice: they deliver known carry income and identify a persistent long-leg core that bears the spot loss. The sample rank correlation between average rate and beta rankings is 0.72. Country size, commodity exposure, trade position, and intermediary demand can all

contribute to that alignment (Hassan, 2013; Ready et al., 2017; Della Corte et al., 2016; Richmond, 2019; Hassan and Mano, 2019; Gabaix and Maggiori, 2015). The analysis does not select a unique primitive; it shows that two predetermined rankings point toward the same loss-bearing currencies.

Table 6: Persistent portfolio cores and state-dependent rotation

Currency	Carry portfolio		Equity-beta portfolio	
	Long	Funding	Long	Funding
New Zealand	85 / 91	2 / 1	77 / 76	0 / 0
Australia	71 / 73	4 / 0	97 / 92	0 / 0
Norway	62 / 34	3 / 5	26 / 32	0 / 0
United Kingdom	43 / 70	4 / 0	0 / 0	52 / 47
Canada	20 / 21	23 / 14	100 / 100	0 / 0
Japan	0 / 0	83 / 97	0 / 0	73 / 65
Switzerland	0 / 0	96 / 100	0 / 0	95 / 85

Notes: Entries are the percentage shares of months in which a currency appears in the indicated three-currency leg, reported as full sample / active state. The full sample contains 458 portfolio months and the active state contains 92. The table selects seven currencies from the complete nine-currency composition table in the Online Appendix.

5.4 How stable is the beta ordering?

Across the reported state-definition neighborhood, the S&P-beta portfolio is more stable than carry. Its spot coefficient is -10.35 under one-month release, -4.23 under three-month release, and -5.25 under the 10Y–3M construction; all three are statistically detected at the reported levels. Removing fitted exposure to the U.S. ACM slope term premium leaves a -6.17 -point beta coefficient with $p = 0.004$, even as the carry coefficient attenuates.

The exposure measure itself has an important failure. Rebuilding the portfolio with expanding MSCI World betas produces a full-sample spot coefficient of -2.93 with $p = 0.319$, compared with -7.93 and $p = 0.005$ for S&P betas on the same maximal sample. After 1999, both definitions are negative and detected. Early Japanese equity weight can explain why an external U.S. index and a capitalization-weighted world index rank the yen differently, but it does not erase the full-sample benchmark dependence.

Table 7: Equity-beta benchmark sensitivity

Sample	Beta benchmark	N	Spot b	Wild p	Total b	Wild p
<i>Panel A: synchronized-inversion state</i>						
Maximal all-nine-beta	S&P 500	443	-7.93	0.005	-7.19	0.009
Maximal all-nine-beta	MSCI World	443	-2.93	0.319	-4.00	0.162
Post-1999	S&P 500	320	-11.10	0.003	-10.73	0.004
Post-1999	MSCI World	320	-7.67	0.030	-7.85	0.019
<i>Panel B: synchronized-inversion state excluding downcurve months</i>						
Maximal all-nine-beta	S&P 500	443	-12.31	0.001	-11.97	0.001
Maximal all-nine-beta	MSCI World	443	-2.30	0.544	-2.65	0.446
Post-1999	S&P 500	320	-13.35	0.001	-13.09	0.001
Post-1999	MSCI World	320	-8.29	0.014	-8.67	0.008

Notes: Annualized percentage-point state coefficients for high-minus-low portfolios sorted each month on expanding currency betas estimated against the named equity index; sorts are lagged one month. Panel A uses each definition's maximal all-nine-beta sample and therefore excludes fifteen early months in which the baseline portfolio uses seven or eight available betas. The full-sample MSCI-based result is not statistically detectable. Japan's declining weight in MSCI World helps explain the post-1999 agreement but does not remove the full-sample benchmark sensitivity.

The state forecasts losses in the high-rate leg, and a predetermined S&P-beta rank orders them across currencies and in a separate portfolio. This benchmark-sensitive gradient turns a time-series gap into a testable statement about who bears it; it is not a structural mechanism.

6 Alternative Explanations and Independent Stress Tests

6.1 The United States, financial conditions, and volatility

The U.S. curve does not substitute for the cross-country configuration. Its inversion alone has a -1.90 -point carry-spot coefficient with $p = 0.484$. Adding it to the configuration count is a different experiment. The resulting ten-curve state contains all 92 baseline months plus nine additional months, and the carry-spot coefficient remains -11.72 ($p = 0.001$). Including the dollar in the portfolio also leaves the estimate near -13 points. The exclusion of the United States is therefore not load-bearing for the reported association, even though the U.S. curve contributes little on its own.

Table 8: Including the U.S. curve and the dollar

Portfolio	Signal construction	Return leg	b	t_{run}	Wild p
G10 carry	Baseline (nine curves)	Spot	-12.85	-3.03	0.001
G10 carry	Baseline (nine curves)	Total	-11.31	-2.40	0.003
G10 carry	Add U.S. curve (ten curves)	Spot	-11.72	-2.92	0.001
G10 carry	Add U.S. curve (ten curves)	Total	-10.20	-2.34	0.006
Dollar-inclusive	Baseline (nine curves)	Spot	-13.11	-2.69	< 0.001
Dollar-inclusive	Baseline (nine curves)	Total	-11.54	-2.15	0.009
Dollar-inclusive	Add U.S. curve (ten curves)	Spot	-12.10	-2.68	0.001
Dollar-inclusive	Add U.S. curve (ten curves)	Total	-10.57	-2.14	0.013

Notes: Coefficients are annualized percentage-point state effects from bare state regressions; t_{run} clusters by signal run and p is a wild bootstrap on signal runs ($B = 9,999$). The baseline state has 92 months. Adding the U.S. curve produces 101 state months and contains all 92 baseline months. The dollar sits in the dollar-inclusive portfolio's funding leg in 31 percent of months. The result does not depend on excluding the U.S. from either margin; this does not make the U.S. curve structurally irrelevant.

Lagged financial conditions do not absorb the baseline state. With the NFCI level and change, the carry-spot coefficient is -12.84 points ($p < 0.001$). Adding lagged VIX level and change gives -13.18 ($p = 0.004$). Contemporaneous dollar and realized FX-volatility controls leave carry spot at -11.5 ($p = 0.002$) and the S&P-beta portfolio at -7.4 ($p = 0.009$). These designs answer different questions. Lagged variables test predictive spanning; contemporaneous variables account for measured features of the realized loss.

The favorable realized-volatility result is not universal. Substituting G10 implied-

volatility innovations leaves the carry estimate statistically detected but produces high-beta coefficients near -7.5 with p -values around 0.17 – 0.18 . The raw current-inversion count is also absorbed by the full NFCI/VIX control set. These failures reinforce the comparative result: the live configuration adds information beyond the displayed lagged conditions, but neither every volatility proxy nor every curve-state construction preserves the same cross-sectional precision.

6.2 Expected rates and a common term-premium proxy

An inversion combines expected short-rate news and term-premium variation. In a predictive-spanning exercise, adding the lagged U.S. ACM slope-premium level, or its level and one-month change, leaves the carry-spot coefficient near -12.9 percent ($p = 0.001$) and the S&P-beta spot coefficient near -7.6 percent ($p = 0.011$). A separate decomposition removes each country’s fitted exposure to the U.S. ACM 10Y-minus-2Y slope term premium, then rebuilds the state. Eighty-three percent of national inversion observations remain. The stripped state is broader—125 months in eighteen runs—and overlaps the baseline in 83 months.

The carry result attenuates: the spot coefficient falls from -12.85 to -5.42 and has $p = 0.107$. The S&P-beta coefficient remains -6.17 with $p = 0.004$. The rebuilt state still contains all five worst carry months but expands from 92 to 125 active months. Its smaller carry coefficient therefore reflects a changed classifier—and possible dilution from the additional months—as well as removal of fitted exposure to the U.S. premium. The exercise does not identify what share of the baseline association is attributable to a common term-premium component, and one U.S. proxy cannot separate expected rates and term premia country by country.

Table 9: State rebuilt after removing fitted exposure to the U.S. ACM slope term premium

State	Carry spot b [p]	Carry total b [p]	High-beta spot b [p]	High-beta total b [p]
Baseline state	−12.85 [0.001]	−11.31 [0.003]	−7.53 [0.010]	−6.82 [0.013]
Stripped state	−5.42 [0.107]	−4.07 [0.251]	−6.17 [0.004]	−5.66 [0.008]
Non-downcurve months	−7.68 [0.094]	−7.03 [0.142]	−10.27 [0.000]	−10.28 [0.000]
Downcurve months	+1.56 [0.621]	+3.81 [0.059]	+5.00 [0.026]	+6.36 [0.012]
Downcurve wedge within stripped state	+7.64 [0.149]	+9.18 [0.094]	+12.89 [0.007]	+14.16 [0.004]

Notes: Sample: 1988:01–2026:02. Each national 10Y–2Y slope is regressed on the U.S. ACM 10Y-minus-2Y term-premium slope; fitted exposure is removed before rebuilding the state with the baseline entry, release, and breadth rules. Cells report annualized coefficients and wild run-bootstrap p -values in brackets ($B = 9,999$). The stripped state has 125 months in 18 runs, versus 92 months in 15 runs for the baseline, and the two share 83 months. Of 1,134 inversion curve-months in the 1980:01–2026:02 slope panel, 942 (83 percent) remain inverted after stripping. This is a fitted adjustment for a common U.S. proxy, not a decomposition of foreign country-specific term premia. The high-beta effect remains statistically detectable; the carry coefficient attenuates and becomes imprecise.

The evidence therefore leaves two live readings. Synchronized curves can summarize coordinated revisions to expected policy and growth, or they can reveal common risk-bearing information embedded in term premia. The currency cross-section travels under the available adjustment; the carry magnitude does not. A consistent international expected-rate/term-premium panel is the data extension most likely to separate these interpretations.

6.3 Calendar, episode influence, and international breadth

Carry spot coefficients are negative in both calendar halves: −11.41 points before 2005 and −15.85 afterward. The S&P-beta portfolio is less stable, with −4.01 ($p = 0.287$) in the first half and −11.91 ($p = 0.010$) in the second. These are recursive historical stability facts, not a holdout, because the state was evaluated over the full history.

Leaving out one episode at a time keeps carry coefficients within roughly one-third of the full-sample value, and the largest reported beta-portfolio bootstrap p -value is 0.006. The exercise does not report every deletion coefficient or a joint leave-two-episode audit. The results are not reducible to one episode, but fifteen episodes remain the effective number of common-state realizations.

Seven floating emerging-market currencies add international breadth without constructing the G10 state. All seven bilateral signs are negative. The pooled spot

coefficient is -14.37 with $p = 0.104$, and the pooled total coefficient is -13.35 with $p = 0.079$. On the common 1996–2026 sample, active-state G10 and G10-plus-emerging-market spot means are similar, -12.79 and -12.51 percent. Total means differ, -8.67 and $+2.51$, because implied interest income is approximately 4.12 versus 15.02 percent. Shared spot repricing need not imply a universally negative total return when predetermined income differs; interpreting that difference as risk compensation requires additional structure.

6.4 An independently executable public 10Y–3M audit

The public audit rebuilds a nearby signal from current-vintage OECD monthly 10-year government yields and 3-month interbank rates distributed through FRED. It pairs that state with a BIS log spot-return proxy ranked on policy-rate differentials available by portfolio formation. The declared 64-rule family crosses raw versus live states, breadth one through four, release lengths one through three, one- and two-month return lags, and target leg sizes of two or three. All currencies tied at a cutoff enter with equal weight, so realized leg size can exceed its target. The code exhaustively reports every rule, but the family was not preregistered before the data were examined.

The baseline-like rule produces a -3.20 -point annualized log spot-return difference across 99 active months and eighteen episodes. Its inclusive circular-shift reference value is 0.335. Forty-one of 64 rules are negative, the baseline ranks 22nd from most negative to most positive, and its common-calendar maximum-standardized-coefficient reference value is 1.000. Negative signs are frequent, but the baseline estimate is imprecise and no rule meets the 5-percent family reference criterion.

Table 10: Independent public 10Y–3M proxy audit

Public-data specification	Log spot difference	Raw rotation ref.	Active months	Episodes
Baseline-like rule	−3.20	0.335	99	18
Exclude episodes containing 1998:09, 2008:10, or 2020:03	−0.46	0.826	88	15
First calendar half, 1988–2004	−5.73	0.382	49	9
Second calendar half, 2005–2025	−1.02	0.776	50	10
Delete five worst active months	+0.94	—	94	—

Notes: Annualized percentage-point active-minus-inactive differences in a BIS policy-ranked log spot-return proxy. Each leg targets three currencies; all currencies tied at a cutoff enter with equal weight, so realized leg size can exceed three. The baseline-like rule uses current-vintage OECD monthly 10-year government yields minus 3-month interbank rates distributed through FRED, fresh entry, breadth two, two consecutive increases for release, and a one-month return lag. It is distinct from the baseline 10Y–2Y state and is not an executable forward return. Across the complete declared family, 41 of 64 coefficients are negative, the baseline ranks 22nd from most negative to most positive, and the common-calendar max- $|z|$ rotation-reference value is 1.000. The crisis-deletion reference rotates the resulting complete-case sequence after excluded episodes are removed. The last row is outcome-conditioned and intentionally has neither a p -value nor an episode count, because deleting realized months mechanically splits state runs.

The public same-universe checks sharpen and limit that conclusion. Jointly deleting each currency from signal construction and the spot basket produces nine negative influence estimates, ranging from −5.08 when Australia is removed to −0.60 when the pound is removed. These estimates share most dates and are not nine independent tests. A geographically disjoint European-curve signal precedes a −15.84-point non-European basket difference across ten observed episodes (raw rotation-reference value 0.062); the reverse direction is −0.34 across nine episodes (reference value 0.887). These auxiliary references rotate each complete-case sequence; internal missing months are compressed rather than treated as literal calendar observations. The two designs also use different target leg sizes and coverage thresholds. Their asymmetry does not validate a universal common factor.

Concentration matters as well. Removing public-proxy episodes containing September 1998, October 2008, or March 2020 attenuates the baseline to −0.46 (complete-case rotation-reference value 0.826). Deleting the five worst active months changes the coefficient to +0.94. The public audit is a novel robustness result because its data, rule

grid, finite-rotation references, episode ledger, and state recursion are independently executable from the saved current-vintage snapshot. It disciplines the central claim sharply: a nearby public configuration often produces the same coefficient sign, but zero rules meet the 5-percent common-calendar family reference criterion and the baseline-like estimate disappears outside the largest losses.

6.5 What the design establishes

The estimates establish a large historical conditional spot-return shift and a nonlinear cross-country configuration. The S&P-based tests document an exposure ordering with the stated benchmark and inferential limits. The U.S., average-slope, own-curve, risk-control, and U.S.-term-premium comparisons narrow nearby explanations. The independent public family shows frequent negative coefficients but no rule meets its 5-percent common-calendar rotation-reference criterion.

Three boundaries remain decisive. The public audit does not regenerate the 10Y–2Y state, its fifteen-episode ledger, or the full specification search behind that state. Money-market differentials are not executable one-month forward returns after basis, spreads, formation timing, and financing costs. Finally, prediction and exposure ordering do not identify the shock or compensation friction. These boundaries define the next empirical agenda; they do not dilute the central economic result.

7 Why Can a Public State Forecast Poor Carry Returns?

7.1 Spot prediction, known income, and total returns

The empirical results separate three objects. First, the state predicts a spot depreciation of the high-rate currencies relative to the funding currencies. That fact is not itself a violation of uncovered interest parity: high-rate currencies can depreciate. Second, the portfolio earns a positive interest differential known at formation. Third, the money-market-differential total-return proxy is negative on average inside the state because the spot movement exceeds that income.

The income leg moves in the compensating direction: it averages 3.60 percent per year outside the state and 5.14 percent inside it, a +1.54-point difference with episode-bootstrap $p = 0.260$. That rise is too small to offset the -12.85 -point spot gap, leaving the -11.31 -point total-return gap. The differential is observed income, not identified risk compensation.

This accounting creates the pricing question. If a public state reveals bad-state covariance, a frictionless asset-pricing benchmark normally assigns compensation before the payoff. A negative conditional mean can still appear in a small sample of rare episodes, or when the measured total return differs from a feasible forward payoff. If it survives those explanations prospectively, prices or positions must adjust slowly enough for current expected losses to exceed compensation already embedded in the differential.

7.2 A benchmark restriction

Let an adverse event of size $J > 0$ occur at $t + 1$ with conditional probability p_t . Write the known income as $C_t = \bar{C}_t + \mu J \mathbb{E}_{t-1}[p_t]$, where $\bar{C}_t$ is ordinary carry and the second term represents compensation based on the earlier information set, with $\mu \geq 1$. Then

$$\mathbb{E}_t[rx_{t+1}] = \bar{C}_t + J \{ \mu \mathbb{E}_{t-1}[p_t] - p_t \}, \quad \mathbb{E}_t[rx_{t+1}] < 0 \iff p_t > \mu \mathbb{E}_{t-1}[p_t] + \frac{\bar{C}_t}{J}. \quad (10)$$

The inequality is the benchmark's only claim. Current perceived loss risk must rise enough to exceed both ordinary income and compensation based on prior beliefs. Fresh synchronized inversions can mark such an information change. The regression does not

observe p_t , estimate J , or identify the lag in compensation.

Three interpretations remain. The conditional mean may be compensation for adverse covariance estimated imprecisely from fifteen episodes. Portfolio adjustment or intermediary constraints may delay the currency response (Bacchetta and van Wincoop, 2010; Gabaix and Maggiori, 2015). Historical rule selection may also make the in-sample conditional mean more negative than its prospective value. These explanations imply different future evidence: executable forward payoffs, position or flow adjustments, and performance under a frozen state.

The paper does not choose among them. Its ambitious, testable implication is upstream: a cross-country bond-market configuration known before the payoff historically predicts losses on the persistent high-rate and high-beta currency exposures. A prospective episode under the dated algorithm can distinguish persistent pricing information from historical classification.

8 Conclusion

The cross-country configuration of yield curves predicts a distinct currency-return state. When fresh 10Y–2Y inversions are live in at least two non-U.S. G10 economies, next-month carry spot returns are 12.9 percentage points per year lower. The known interest differential absorbs only part of that depreciation. The opposite signs in the separate exactly-one and exactly-two indicator regressions, together with the portfolio horse race, distinguish the configuration from inversion in general, the U.S. curve, and the reported continuous average-slope and own-state measures. Those indicator coefficients are each measured against a different complement and are not a direct one-versus-two contrast.

The cross-section gives that forecast economic content. High-rate currencies lose relative to low-rate funding currencies, and bilateral losses grow with predetermined S&P equity beta. Persistent portfolio cores show that the result is not created solely by rotation at signal dates. The S&P-based exposure ordering is more stable across the reported tenor, release, and common-term-premium perturbations than the precise carry magnitude, although the nine-point fit is descriptive and the full-sample beta result depends on the equity benchmark.

An independently executable public 10Y–3M audit frequently produces a negative coefficient across its declared family, but no rule meets the 5-percent common-calendar rotation-reference criterion and the result attenuates after crisis-episode deletion. The public proxy therefore does not independently confirm the baseline relation. It shows why the exact carry coefficient cannot be treated as invariant to tenor, state persistence, or outcome construction. The main empirical object is the baseline configuration and the exposure-ordered loss it precedes.

The next decisive evidence is now well defined. Freeze the 10Y–2Y algorithm and vintage protocol, release the complete country-month state matrix, test each currency with its own curve excluded, and measure executable forward payoffs in the next unused episode. Those tests can determine whether the configuration is a durable international signal and whether its spot forecast translates into a negative conditional carry payoff.

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